Temperature-dependent frequency control of TWC operation for the exhaust gas aftertreatment of stoichiometric natural gas engines

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1. Introduction

Reducing CO_2 emissions remains a critical objective in global climate change mitigation strategies. As the U.S. transport sector accounts for nearly 40% of energy-related CO_2 emissions [1], it represents a key target for significant emission reductions. The use of methane (CH_4) offers a promising reduction in CO_2 emissions due to its higher hydrogen-to-carbon ratio (H:C = 4:1) in comparison to conventional fuels. Through power-to-gas (PtG) technologies, methane can be synthesized using renewable energy sources, allowing it to serve not only as an interim solution but also as a prospective long-term, carbon-neutral energy carrier in sustainable mobility systems. Nevertheless, the reduction of methane emissions at low temperatures and the avoidance of secondary emissions, such as N_2O and NH_3, remain significant challenges in the context of three-way catalysts (TWC). As previously shown, switching between a lean and rich exhaust gas mixture at a temperature-dependent optimal frequency can maximize pollutant conversion and N_2 selectivity [2]. To utilize the relationship between optimal frequency and exhaust gas temperature during catalyst cold start we designed and tested a temperature-dependent frequency control system for a Pt-based TWC using a realistic synthetic exhaust gas representative for a typical stoichiometric natural gas engine.

2. Methodology

A temperature-frequency sweep was conducted on a monolithic Pt/Al_2O_3 catalyst to ascertain the relationship between inlet temperature and optimal frequency, employing a polynomial expression. This expression was then utilized to adjust the frequency during light-off measurements.

3. Results and discussion

Figure 1 shows the CO, NO and CH_4 conversion, the N_2 selectivity as well as N_2O, NH_3 and NO_2 emissions during catalyst cold start (150–600 °C, 5 K min^{-1}) at conventional (λ = 1) and fully-mixed ($f \rightarrow \infty$) stochiometric steady-state condition and at periodic operation at constant (1 Hz, 1.5 Hz and 2 Hz) and temperature dependent

frequency ($f = f_{opt}(T)$) at $A = 0.04$ and $GHSV = 75\,000\ h^{-1}$. As the desorption rate increases with rising temperature, higher frequencies are necessary to prevent the poisoning of the catalytic surface by the most abundant surface species under the prevailing conditions [2,3]. Therefore, controlling the frequency of the rich-lean switches with the exhaust gas temperature results in a decrease of the T_{50} light-off temperature of NO_x and CH_4 by 210 K and 120 K, respectively and enhance the N_2 selectivity compared to conventional stoichiometric steady-state operation. On the other hand, periodic operation at a constant frequency results in an enhancement of NO_x and CH_4 conversion only in a specific temperature range. Furthermore, operating the catalyst at a constant frequency increases high-temperature NH_3 emissions, which can be attributed to the formation of H_2 from steam reforming of CH_4 at rich conditions. The present work proposes a cost-effective and simple method for optimizing TWC operation during catalyst cold start through the utilization of periodic operation.

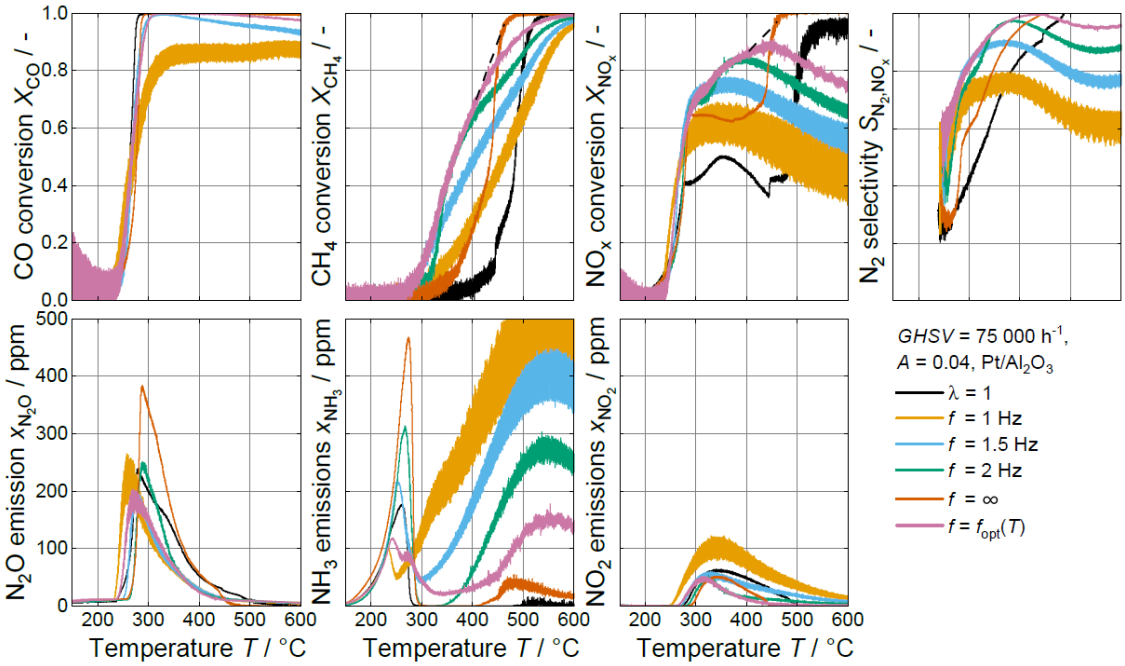

Figure 1. CO, CH_4, NO_x conversion and N_2 selectivity as well as N_2O, NH_3 and NO_2 emission during light-off experiments (5 K min^{-1}) at conventional stoichiometric steady state (λ = 1), 1 Hz, 1.5 Hz, 2 Hz, fully mixed stoichiometric steady state (f → ∞) and temperature dependent frequency control ($f = f_{opt}(T)$) on Pt/Al_2O_3 at constant GHSV = 75 000 h^{-1} and amplitude A = 0.04. (Gas composition: 17% H_2O, 9% CO_2, 800 ppm CH_4 and either 0.74 % O_2, 4000 ppm NO, 900 ppm CO, 300 ppm H_2 (lean feed), 0.13% O_2, 1000 ppm NO, 1.17% CO, 0.39% H_2 (rich feed) or 0.28% O_2, 1600 ppm NO, 0.31% CO, 0.10% H_2 (*λ = 1*) [3].

References

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